SYARIF ADRIAN MANGARAJA LUBIS

Information Systems Undergraduate

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SUMMARY

Information Information Systems student with a 3.79 GPA and a strong focus on applied artificial intelligence and machine learning. Highly proficient in utilizing AI-assisted development workflows (agent orchestration, code review, and prompt engineering) to optimize software delivery and system design. Hands-on experience developing predictive models, data pipelines, and web backends, highlighted by a production-level apron management and recommendation system deployed in an airport environment

PROJECTS

ML-Based Aircraft Parking Stand Recommendation System -

Undergraduate Thesis - Suryadarma Aerospace University

Multi-class recommendation engine that suggests optimal parking stand assignments for the Apron Movement Control unit at Halim Perdanakusuma International Airport, deployed as a web application currently in UAT.

- Stack : Python, Scikit-learn, PHP MVC (Custom)
- Built and trained a Random Forest classifier on 5,190 historical aircraft movement records (May–July 2025) across 17 parking stand classes, using aircraft type, airline, and flight category as features.
- Achieved 80.15% Top-3 accuracy, the primary evaluation metric given the system's design as a three-option recommender supporting controller judgment rather than an automated decision-maker.
- Addressed severe class imbalance across stands by applying SMOTE oversampling and class weighting, and documented the trade-offs between Top-1 (36.32%) and Top-3 performance in a 17-class problem
- Integrated the trained model into a PHP MVC application with a live apron view, delivering recommendations to AMC officers in real time.

Airline Sentiment Analysis Using Machine Learning

Data Mining Course Project – Suryadarma Aerospace University

- Stack : Python, Scikit-learn, NLTK, Pandas, NumPy, Matplotlib, Seaborn
- Sentiment classifier for airline customer reviews sourced from Twitter, labelling feedback as positive, neutral, or negative.
- Preprocessed raw text data and applied TF-IDF vectorization for feature extraction.
- Compared LinearSVC and Complement Naive Bayes; LinearSVC achieved 76.3% accuracy and a 76.3% weighted F1-score.
- Conducted exploratory data analysis to surface recurring themes in customer feedback, including service quality issues and flight delays.

EDUCATION

Undergraduate of Information Systems
2022 - Present

Suryadarma Aerospace University

- GPA of 3.79 / 4.00 (Final)
- Relevant Coursework: Data Mining, Business Intelligence, Graphics Design, Internet Technology, Multimedia, Application Programs, HTML, SQL, Database Systems, Operating Systems
- Contributed to the HIMIPEDIA E-commerce Website Faculty Project as UI/UX Designer

Computer and Network Engineering
2019 - 2022

Bisnis Informatika Vocational High School

- Hands On Experience in Computer and Network Hardware
- Ranked #1 in class for 4 consecutive semesters

WORK EXPERIENCE

Apron Movement Control Staff
May 2025 - August 2025

Halim Perdanakusuma International Airport

- Managed 82 parking stands across 3 apron areas at HLP International Airport, ensuring optimal aircraft positioning
- Responsible for operational coordination in the apron area, including real-time aircraft parking stand allocation and movement monitoring
- Coordinated operational communications with airlines, ground handling teams, and technical departments via radio
- Documented aircraft movement data and participated in weekly FOD Walk safety activities
- Developed a proprietary apron management system featuring live apron view and ML-based parking stand recommendation, currently in User Acceptance Testing (UAT) phase

Officer In Charge Staff
March 2025 - May 2025

Halim Perdanakusuma International Airport

- Managed visitor pass applications and approvals for business partners, commercial tenants, and internal airport units, ensuring compliance with security procedures
- Reviewed and approved operational permits including cargo area access, airside vehicle permits, and special activity authorizations
- Administered the charter flight slot website, verifying slot applications, updating availability status, and providing technical support to operators
- Compiled operational and administrative documents including daily reports, meeting minutes, official correspondence, and coordination documentation
- Developed a prototype airport facilities information website featuring a user-friendly interface, analytics features, and a content management dashboard

Computer Lab Assistant
September 2025 - December 2025

Yayasan Titian Ilmu Bekasi | Rawalumbu, Bekasi

- In charge of the entire foundation's network and technology management
- Kept inventory, recorded, and quality-checked lab equipment
- Prepared the lab for student sessions, ensuring hardware and software readiness
- Installed and expanded wired and wireless internet networks across 3 floors of 2 buildings for the foundation

Apprentice Technician

June 2021 – September 2021

CV. Indo Jaya Teknik | Cipendawa, Bekasi

- Assisted in repairs, installations, maintenance, and mapping of digital security devices under a senior technician's supervision
- Drafted administrative documents including handover reports and event records
- Contributed to the expansion of the digital security network across multiple branches of Bank CIMB Niaga

ORGANIZATIONAL EXPERIENCE

Lead Photographer and Graphic Designer

Informatics Student Association, Suryadarma Aerospace University

- Led photography and videography for all organization events during tenure
- Designed posters, frames, twibbons, stickers, and other graphics for documentation, publication, and souvenirs
- Managed the organization's social media account

ADDITIONAL INFORMATION

Soft Skills: Public Speaking, Teamwork, Time Management, Adaptability, Leadership

Technical Skills: Python (Scikit-learn, Pandas, NumPy, NLTK, Matplotlib, Seaborn), Machine Learning, NLP, AI-Assisted Development (Claude Code agent orchestration, prompt engineering, code review & debugging), SQL & Database Systems, PHP (MVC), HTML, Git, Figma, Adobe Photoshop, Canva, Microsoft Office, Google Suite

Languages: English, Indonesian

Certifications: TOEFL, Bright Academy. Score: 663 | Claude Code in Action, Anthropic